

## Chemistry, 2017 EC, natural science, entrance exam, round 1

Grade 12 · Chemistry · 2017 E.C.

Time allowed: 120 minutes

Questions: 72

Total marks: 72

Name \_\_\_\_\_ Grade \_\_\_\_\_ ID No. \_\_\_\_\_ Date \_\_\_\_\_

1. Which type of glass is made by heating a mixture of silica, sodium carbonate and limestone? (1)
  - A. Pyrex glass
  - B. Soda-lime
  - C. Borosilicate
  - D. Quartz
2. Consider the following galvanic cell: When the Zn strip was connected to metal 'B' at 25°C, the voltmeter shows 0.51 V. Which of the following is the standard reduction potential of metal 'B'? ( $E_{\text{Zn}^{2+}/\text{Zn}}^{\circ} = -0.76 \text{ V}$ ) (1)
  - A.  $E_{\text{B}^{2+}/\text{B}}^{\circ} = 1.27 \text{ V}$
  - B.  $E_{\text{B}^{2+}/\text{B}}^{\circ} = 0.25 \text{ V}$
  - C.  $E_{\text{B}^{2+}/\text{B}}^{\circ} = -1.27 \text{ V}$
  - D.  $E_{\text{B}^{2+}/\text{B}}^{\circ} = -0.25 \text{ V}$
3. Which of the following polymer is used to make ropes, clothes, hair combs and stockings? (1)
  - A. Bakelite
  - B. Teflon
  - C. Nylon
  - D. Perspex
4. Which of the following catalyst is used in the Ostwald process for the production of  $\text{HNO}_3$ ? (1)
  - A. CuO
  - B. Fe
  - C.  $\text{V}_2\text{O}_5$
  - D. Pt
5. Which of the following factors does NOT affect the rate of a chemical reaction having solid reactants? (1)
  - A. Volume of reaction vessel
  - B. Physical state of reactants
  - C. Temperature of reaction
  - D. Concentration of reactants
6. What is the electron configuration of  $\text{O}_2^-$  molecule as per the molecular orbital model? (Atomic number of O = 8) (1)
  - A.  $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \sigma_{2px}^2 (\pi_{2py}^2 = \pi_{2pz}^2) (\pi_{2py}^{*1} = \pi_{2pz}^{*1})$
  - B.  $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \sigma_{2px}^2 (\pi_{2py}^2 = \pi_{2pz}^2) (\pi_{2pz}^{*2})$
  - C.  $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \sigma_{2px}^2 (\pi_{2py}^2 = \pi_{2pz}^2) (\pi_{2py}^{*2} = \pi_{2pz}^{*2})$
  - D.  $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \sigma_{2px}^2 (\pi_{2py}^2 = \pi_{2pz}^2) (\pi_{2py}^{*2} = \pi_{2pz}^{*1})$
7. At a particular temperature, the equilibrium constant,  $K_c = 0.36$  for the reaction  $\text{SO}_3(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \text{O}_2(\text{g})$ . Which of the following predictions can be drawn from the given data? (1)
  - A. It is not possible to predict the direction of the reaction.
  - B. The reaction will proceed to the right.
  - C. The reaction will proceed to the left.
  - D. The reaction is at equilibrium.
8. A mineral absorbs purple light of frequency  $7.11 \times 10^{14} \text{ Hz}$ . What is the wavelength (in nm) of the absorbed light? ( $c = 3.00 \times 10^8 \text{ m/s}$ ) (1)
  - A. 314.5 nm
  - B. 421.9 nm
  - C. 237.5 nm
  - D. 184.4 nm

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9. Why do atoms absorb energy when their electrons undergo transitions from a lower energy to a higher energy level? This is because (1)
- A. The electrons of the atoms absorb energy when they go from higher to lower energy.
  - B. The electrons of the atoms do not have allowable energy levels.
  - C. The electrons of the atoms move from one orbit of higher radius to another one having lower radius.
  - D. The electrons of the atoms move from one orbit of lower radius to another one having larger radius.
- 
10. What are the quantum numbers for the 5s orbital? (1)
- A.  $n = 5, l = 0, m_l = 0, m_s = \pm \frac{1}{2}$
  - B.  $n = 5, l = 2, m_l = -2, m_s = \pm \frac{1}{2}$
  - C.  $n = 5, l = 1, m_l = -2, m_s = \pm \frac{1}{2}$
  - D.  $n = 5, l = 0, m_l = -1, m_s = \pm \frac{1}{2}$
- 
11. Which of the following is CORRECT about the set of quantum numbers assigned to the two electrons of the helium atom? (1)
- A. The magnetic quantum numbers of both electrons are different.
  - B. The second electron occupies the same orbital as the first with the opposite spin quantum number.
  - C. The second electron occupies the same orbital as the first with the same spin quantum number.
  - D. Both electrons have the same four quantum numbers.
- 
12. What is the bond formed between two or more atoms by sharing of electrons? (1)
- A. Valence bond
  - B. Covalent bond
  - C. Metallic bond
  - D. Ionic bond
- 
13. Which statement below CORRECTLY describes dipole-dipole forces? It is the force that exists (1)
- A. between polar and non-polar molecules.
  - B. between non-polar molecules.
  - C. between polar molecules.
  - D. in all
- 
14. Comparing the forces between molecules of similar molar masses, which one of the following produces the weakest force? (1)
- A. Van der Waals forces.
  - B. Ionic forces.
  - C. Hydrogen bonding.
  - D. Covalent bonds
- 
15. Which of the following is NOT true about greenhouse effect and greenhouse gases? (1)
- A. The greater the percentage of greenhouse gases in the atmosphere, the cooler the earth should be.
  - B. Oxygen (O<sub>2</sub>) and nitrogen (N<sub>2</sub>) are transparent to infrared radiation.
  - C. Carbon dioxide and water vapor are greenhouse gases.
  - D. Greenhouse effect is the trapping of the infrared radiation by certain gases in the atmosphere
- 
16. In which of the following is the type of natural resource and the given example CORRECTLY paired? (1)
- A. Non-Renewable Resources; water
  - B. Non-Renewable Resources; animal wool
  - C. Renewable resources; crops
  - D. Renewable resources; copper
- 
17. Which of the following compound is a Bronsted-Lowry base? (1)
- A.  $\text{H}_2\text{S}$
  - B.  $\text{HCl}$
  - C.  $\text{NH}_3$
  - D.  $\text{BF}_3$

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18. In an activity during laboratory session a small amount of lead bromide ( $\text{PbBr}_2$ ) crystals were placed in a beaker. Two electrodes are inserted until they are in contact with the  $\text{PbBr}_2$  crystals. When the  $\text{PbBr}_2$  crystals in the beaker are gently heated, which of the following will be observed? (1)
- A. Lead (II) ion is oxidized at the cathode  
B. Lead (II) ion is reduced at the anode  
C. Bromide ion is oxidized at the cathode  
D. Bromine gas is evolved at the anode
- 
19. What is the name given to the pollution caused by the dumping of non-biodegradable wastes into the environment? (1)
- A. Water pollution  
B. Sound pollution  
C. Land pollution  
D. Air pollution
- 
20. What is the purpose of the conversion of pig iron to steel? (1)
- A. To decrease the concentration of iron in the pig iron  
B. To increase the concentration of iron in the pig iron  
C. To remove impurities by reduction  
D. To remove impurities by oxidation
- 
21. Which of the following statements CORRECTLY describes the law of conservation of mass? (1)
- A. Pure compounds always contain the elements in the same percentage by mass.  
B. A particular compound is composed of the same elements in the same parts by mass.  
C. Mass is neither created nor destroyed during a chemical reaction.  
D. The total mass of substances varies during a chemical reaction.
- 
22. What is the density of methane ( $\text{CH}_4$ ) at a pressure of 900 torr and a temperature of  $25^\circ\text{C}$ ? ( $R = 0.082 \text{ atm. L/mol. K}$ ,  $1 \text{ atm} = 760 \text{ torr}$ ) (1)
- A. 9.2  
B. 1.25  
C. 0.92  
D. 0.78
- 
23. It takes 2.25 minutes for 0.02 mol of He to diffuse. How long will it take to diffuse for the same amount of methane gas,  $\text{CH}_4$ ? (M.wt of He = 4 g/mol,  $\text{CH}_4$  = 16 g/mol) (1)
- A. 6 min  
B. 4.5 minutes  
C. 2.25 min  
D. 1.25 min
- 
24. Which one of the following statement CORRECTLY defines rate of reaction? The rate of a reaction is (1)
- A. the measure of the rate of diffusion of different gases inside a tube.  
B. the study of necessary mechanisms occurred in a chemical reaction.  
C. the measure of the amount of energy change in a chemical reaction.  
D. the changes in concentration of reactants or products per unit time.
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25. Consider the following equilibrium:  $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \Delta H = -198.2 \text{ kJ}$ . Which of the following changes will shift the equilibrium to the left? (1)
- A. Decreasing the concentrations of the reactants and increasing the concentration of the product.  
B. Increasing the concentrations of the reactants and decreasing the concentration of the product.  
C. Increasing the concentration of the reactants and products equivalently.  
D. Decreasing the concentrations of the reactants and products equivalently.

26. Aluminum reacts with oxygen to form aluminum oxide. What is the coefficient of aluminum after balancing the reaction? (1)
- A. 8  
B. 6  
C. 4  
D. 2
27. What property of an ionic compound can be investigated by heating it in a crucible? (1)
- A. Crystallization point  
B. Malleability  
C. Solubility  
D. Melting point
28. Silver has two known isotopes,  $^{107}\text{Ag}$  and  $^{108}\text{Ag}$  with their percent abundance of 52% and 48%, respectively. What is the average atomic mass of silver? (1)
- A. 108.86  
B. 107.48  
C. 108.06  
D. 107.89
29. Which reaction scheme below illustrates the formation of a cation? (1)
- A.  $\text{Mg}^{2+} + 2e^{-} \rightarrow \text{Mg}$   
B.  $\text{S} + 2e^{-} \rightarrow \text{S}^{2-}$   
C.  $2\text{Cl}^{-} \rightarrow \text{Cl}_2 + 2e^{-}$   
D.  $\text{Na} \rightarrow \text{Na}^{+} + e^{-}$
30. Which statement distinguishes equilibrium point from end point in acid-base titration? (1)
- A. End point is a point at which the indicator changes color whereas equivalence point is the point where an acid has completely reacted with a base.  
B. Equivalence point is a point at which the indicator changes color whereas end point is the point where an acid has completely reacted with a base.  
C. Equivalence point is a point at which reactants start to melt whereas end point is the point where products start to freeze.  
D. Equivalence point is a point at which reactants start to evaporate whereas end point is the point where a product starts to condense
31. Which one is the mathematical expression for Faraday's first law of electrolysis? (1)
- A.  $m = \frac{MIt}{nF}$   
B.  $m = \frac{MPV}{RT}$   
C.  $M = \frac{n}{V(\text{liter})}$   
D.  $\frac{m_1}{m_2} = \frac{E_1}{E_2}$
32. Which of the following describes the process of manufacturing valuable products in industries? (1)
- A. Manufacturing industry uses only inorganic chemicals in the production of valuable products.  
B. Carry out unwanted reaction of a material that result in the dissolution or consumption of the material.  
C. Discharging of a solid, liquid or gaseous substance into an environment that causes unwanted changes.  
D. Designing to produce a desired output from raw materials using energy through different steps.
33. For the dissociation of  $\text{HNO}_2$ :  $\text{HNO}_2(aq) \rightleftharpoons \text{NO}_2^{-}(aq) + \text{H}^{+}(aq)$  What is the percent ionization of a 0.5 M solution of  $\text{HNO}_2$ ? ( $K_a$  of  $\text{HNO}_2 = 7.1 \times 10^{-4}$ ) (1)
- A. 9.1%  
B. 3.8%  
C. 1.8%  
D. 0.2%
34. Which of the following reaction shows the action of  $\text{H}_2\text{SO}_4$  as oxidizing agent? (1)
- A.  $\text{Mg}(s) + 2\text{H}_2\text{SO}_4(aq) \rightarrow 3\text{MgSO}_4(aq) + (\text{NH}_2)_2\text{SO}_4(aq)$   
B.  $\text{Cu}(s) + 2\text{H}_2\text{SO}_4(aq) \rightarrow \text{CuSO}_4(aq) + \text{SO}_2(g) + 2\text{H}_2\text{O}(l)$   
C.  $2\text{KOH}(aq) + \text{H}_2\text{SO}_4(aq) \rightarrow \text{K}_2\text{SO}_4(aq) + 2\text{H}_2\text{O}(l)$   
D.  $\text{SO}_3(g) + \text{H}_2\text{O}(l) \rightarrow \text{H}_2\text{SO}_4(aq)$

35. A 0.47 M aqueous solution of a weak base has a pOH of 2. What is  $K_b$  for the weak base? (1)
- A.  $4.4 \times 10^{-1}$                       B.  $1.0 \times 10^{-2}$   
C.  $3.14 \times 10^{-3}$                       D.  $2.17 \times 10^{-4}$
- 
36. A student was given a standard  $\text{Zn(s)}|\text{Zn}^{2+}(\text{aq})$  half-cell and another half-cell containing copper immersed in 1.00 M  $\text{CuSO}_4(\text{aq})$ . When the Zn was connected as the anode at 25°C, the cell potential reading on the voltmeter shows 1.1 V. Which of the following is NOT correct regarding the cell potential measurement? (1)
- A. Electrons travel through the external circuit from copper cathode to the zinc anode.  
B. The zinc electrode was dissolved and the blue color  $\text{CuSO}_4$  solution would fade.  
C. The circuit is completed inside the cell by migration of ions through the salt bridge.  
D. The mass of copper was increased during the cell potential measurement.
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37. Which of the following properties is the property of an acidic oxide? An acidic oxide reacts with (1)
- A. Water to form acids.  
B. Salts to form acids.  
C. Basic oxides to form hydroxides and water.  
D. Bases to form basic hydroxide and water.
- 
38. What is the name given to an acid that dissociates (ionizes) only to a slight extent in aqueous solution? (1)
- A. Strong acid                      B. Dilute acid  
C. Concentrated acid                      D. Weak acid
- 
39. Solutions can be classified based upon their pH values. What is a solution whose pH value is 13? (1)
- A. Acidic solution                      B. Amphoteric solution  
C. Basic solution                      D. Neutral solution
- 
40. During laboratory class, a student placed clean water in a beaker and added pieces of calcium metal using tongs. After the reaction is completed, he/she tested the solution using red litmus paper. Which of the following will be observed? (1)
- A. The litmus paper turns to red.                      B. The litmus paper turns to blue.  
C. The litmus paper turns to colorless.                      D. The litmus paper maintains its color.
- 
41. Which of the following statements CORRECTLY describes the relationship among wavelength, frequency and speed in electromagnetic radiation? (1)
- A. As the wavelength decreases, the frequency increases.  
B. As the wavelength increases, the frequency increases.  
C. As the frequency increases, the speed of light decreases.  
D. As the wavelength decreases, the speed of light increases.
- 
42. Which of the following organic compounds has the highest boiling point? (1)
- A. Ethene                      B. Ethanol  
C. Ethane                      D. Ethanal
- 
43. What is a Lewis base? It is (1)
- A. Proton acceptor                      B. Proton donor  
C. Electron pair donor                      D. Electron pair acceptor

44. Given the following information:  $\text{CH}_3\text{COOH (aq)} \rightleftharpoons \text{CH}_3\text{COO}^- \text{(aq)} + \text{H}^+ \text{(aq)}$   $\text{CH}_3\text{COOH}$   $\text{CH}_3\text{COO}^-$   $\text{H}^+$  Initial concentration (mol/L) 0.2 0 0 Change -X +X +X Equilibrium concentration 0.2 - X X X Which of the following expressions represents the percent ionization of  $\text{CH}_3\text{COOH}$ ? (1)
- A.  $(x/0.2)100\%$  B.  $((0.2-x)/0.2)100\%$   
C.  $(x/(0.2-x))100\%$  D.  $((0.2)/x)100\%$
45. Scandium (Sc) has the electron configuration  $[\text{Ar}] 4s^2 3d^1$ . To which group of elements does it belong to? (1)
- A. Transition elements B. Representative elements  
C. Nonmetals D. Inner transition elements
46. A student carried out an experiment to determine the melting points of  $\text{NaCl}$  and  $\text{CuCl}_2$ . He/she added 0.2 g of  $\text{NaCl}$  and  $\text{CuCl}_2$  into two different test tubes and heated both tubes simultaneously using a Bunsen burner. Which of the following statements is CORRECT regarding the melting point of  $\text{NaCl}$  and  $\text{CuCl}_2$ ? (1)
- A. The melting point of  $\text{NaCl}$  is higher than that of  $\text{CuCl}_2$ .  
B. The melting point of  $\text{NaCl}$  is lower than that of  $\text{CuCl}_2$ .  
C. Both  $\text{NaCl}$  and  $\text{CuCl}_2$  will not melt.  
D. Both  $\text{NaCl}$  and  $\text{CuCl}_2$  will melt at the same time.
47. What is the cause of unusual high boiling points of  $\text{HF}$ ,  $\text{H}_2\text{O}$  and  $\text{NH}_3$ ? (1)
- A. Induced dipole forces B. Dipole-Dipole forces  
C. Ion-dipole forces D. Hydrogen bonding
48. Which statement below describes CORRECTLY the assumptions in the kinetic theory of gases? (1)
- A. Gas molecules are in random motion with no interactions.  
B. Gas molecules occupy a finite volume and have indefinite shape.  
C. The average kinetic energies of gas molecules are independent of the amount of gas.  
D. The average kinetic energies of gas molecules are independent of temperature.
49. If the rate of diffusion of a certain unknown gas is half times the rate of diffusion of helium, what will be the molar mass of the unknown gas? (Atomic mass of  $\text{He}$ = 4.0 g/mole) (1)
- A. 8.0 g/mole B. 2.0 g/mole  
C. 32.0 g/mole D. 16.0 g/mole
50. A student added boiling chips into a beaker of water while carrying out an activity to determine the boiling point of water. Why do you think the boiling chips are necessary? (1)
- A. To speed up the rate of evaporation  
B. To ensure smooth rate of evaporation  
C. To reach the boiling point with less consumption of heat  
D. To maintain constant temperature and pressure
51. What is the oxidation number of manganese in  $\text{KMnO}_4$ ? (1)
- A. +6 B. +5  
C. +8 D. +7
52. 2-methyl-1-butene is one of the isomers of  $\text{C}_5\text{H}_{10}$ . Which of the following is the CORRECT structural formula of this isomer? (1)
- A.  $\text{CH}_3\text{-CH}_2\text{-C(CH}_3\text{)=CH}_2$  B.  $\text{CH}_3\text{-CH(CH}_3\text{)-CH=CH}_2$   
C.  $\text{CH}_3\text{-CH=CH-CH}_2\text{-CH}_3$  D.  $\text{CH}_3\text{-C(CH}_3\text{)=CH-CH}_3$

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53. Which of the following organic compounds is used for the production of 1, 2-ethanediol? (1)
- A. Ethane  
B. Ethene  
C. Ethanol  
D. Ethyne
- 
54. Which of the following is the main constituent of natural gas? (1)
- A. Butane  
B. Ethane  
C. Methane  
D. Propane
- 
55. In order to avoid the decrease in quality of some clothes after washing with water, which of the following chemicals is used for dry cleaning? (1)
- A.  $\text{CH}_2\text{Cl}_2$   
B.  $\text{CH}_4$   
C.  $\text{C}_2\text{Cl}_4$   
D.  $\text{C}_2\text{H}_4$
- 
56. The molecule of  $\text{BF}_3$  has both bonding and lone pair of electrons. Based upon these electrons, which one is the CORRECT molecular geometry of the molecule of  $\text{BF}_3$ ? (1)
- A. Square pyramidal  
B. Square planar  
C. Trigonal bipyramidal  
D. Trigonal planar
- 
57. Which law governs the fact that the ratio of the amount of hydrogen to oxygen by weight will be the same in a drop of water and a lake of water? (1)
- A. Law of conservation of mass  
B. Law of relative composition  
C. Law of definite proportion  
D. Law of multiple proportions
- 
58. Which statement below CORRECTLY describes the characteristics of chemical equilibrium? (1)
- A. A reaction stops if sufficient amount of time is given.  
B. A reaction stops when it reaches equilibrium.  
C. A new substance or substances are formed when equilibrium is attained.  
D. The rates of the forward and reverse reactions are equal.
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59. What is the name given to compounds of carbon and hydrogen containing multiple bonds? (1)
- A. Aromatic hydrocarbons  
B. Aliphatic hydrocarbons  
C. Unsaturated hydrocarbons  
D. Saturated hydrocarbons
- 
60. The following reaction can be used for the synthesis of  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$  as a desired product:  $\text{CH}_3\text{CH}_2\text{CHBrCH}_3 + \text{CH}_3\text{S}^- \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3 + \text{CH}_3\text{SH} + \text{Br}^-$  Which of the following is the atom economy for the above reaction? (Atomic masses: H=1; C=12; S=32 and Br=80) (1)
- A. 48%  
B. 30.43%  
C. 80%  
D. 56%
- 
61. Which of the following process involves preventing the growth of bacteria, fungi and reducing the oxidation of fats? (1)
- A. Tanning  
B. Haber-Bosch  
C. Contact  
D. Food preservation
- 
62. Which of the following is the overall reaction for electrolysis of molten  $\text{NaCl}$ ? (1)
- A.  $\text{Na}^+(\text{l}) + \text{Cl}^-(\text{l}) \rightarrow \text{Na}(\text{l}) + \frac{1}{2}\text{Cl}_2(\text{g})$   
B.  $\text{Na}^+(\text{l}) + \frac{1}{2}\text{Cl}_2(\text{g}) \rightarrow \text{Na}^+(\text{l}) + \text{Cl}^-(\text{l})$   
C.  $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g})$   
D.  $\text{H}_2\text{O}(\text{g}) \rightarrow \text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$

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63. Which of the following occur during the electrolysis of dilute  $\text{H}_2\text{SO}_4$ ? (1)
- A. The overall reaction is:  $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g})$
  - B.  $\text{SO}_4^{2-}$  is discharged at the anode.
  - C. A basic solution is formed
  - D.  $\text{H}^+$  is discharged at the negative electrode
- 
64. Which statement below CORRECTLY distinguishes redox and non-redox reactions? (1)
- A. Reduction occurs in redox reactions and oxidation takes place in non-redox reactions.
  - B. Electrons are transferred in redox reactions but not in non-redox reactions.
  - C. Acid-base reactions can be considered as both redox and non-redox reactions.
  - D. Oxidation number remains the same in both redox and non-redox reactions
- 
65. Given the reaction:  $\text{C}(\text{s}) + \text{CO}_2(\text{g}) \rightleftharpoons 2\text{CO}(\text{g})$  What is the equilibrium constant expression for the reaction? (1)
- A.  $[\text{CO}]^2/[\text{CO}_2]$
  - B.  $[\text{CO}]^2/[\text{C}][\text{CO}_2]$
  - C.  $[\text{C}][\text{CO}_2]/[\text{CO}]^2$
  - D.  $[\text{CO}_2]^2/[\text{C}]$
- 
66. Which of the following pair of substances can be considered as examples of the liquid state of matter at room temperature and 1 atm pressure? (1)
- A. Water and carbon dioxide
  - B. Carbon and mercury
  - C. Carbon and carbon monoxide
  - D. Water and mercury
- 
67. Which gas law describes the behavior of gases using the variables: temperature, volume, pressure and number of moles? (1)
- A. Boyle's law
  - B. Ideal gas law
  - C. Combined gas law
  - D. Avogadro's law
- 
68. Which one of the following CORRECTLY shows the relative charges of an electron, proton and neutron, respectively? (1)
- A. 0, -1, +1
  - B. 0, +1, -1
  - C. -1, 0, +1
  - D. -1, +1, 0
- 
69. Which statement below CORRECTLY describes the Bohr's model of an atom? (1)
- A. Electrons and neutrons are found in the nucleus.
  - B. Protons and electrons are found in the nucleus.
  - C. Electrons move in circular orbits around the nucleus.
  - D. Protons move in circular orbits around the nucleus.
- 
70. Which statement below CORRECTLY describes periodicity? Periodicity is (1)
- A. a row in the Periodic Table which contains elements with same number of shells.
  - B. a regular repetition of chemical and physical properties in the Periodic Table.
  - C. a column in the Periodic Table which contains elements with similar chemical properties.
  - D. a measure of electronegativity which runs from least electronegative to most electronegative elements.

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- 71.** Which statement below CORRECTLY expresses hydrogen bonding? It is formed by bonding (1)
- A. a hydrogen atom to an element such as oxygen.
  - B. a hydrogen atom to an element such as sodium.
  - C. two hydrogen atoms with unequal sharing of electrons.
  - D. two hydrogen atoms with equal sharing of electrons.
- 
- 72.** Consider the molecular structure of  $\text{COCl}_2$  shown below. What are the predicted bond angles  $\text{Cl-C-O}$  and  $\text{Cl-C-Cl}$ ? (1)
- A.  $\text{Cl-C-O} < 120^\circ$  and  $\text{Cl-C-Cl} = 120^\circ$
  - B.  $\text{Cl-C-O}$  and  $\text{Cl-C-Cl} = 120^\circ$
  - C.  $\text{Cl-C-O} > 120^\circ$  and  $\text{Cl-C-Cl} < 120^\circ$
  - D.  $\text{Cl-C-O} = 120^\circ$  and  $\text{Cl-C-Cl} > 120^\circ$

— END OF EXAMINATION —

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